

Notes: Rabin and Schrag (QJE, 1999)
First Impressions Matter: A Model of Confirmatory Bias

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1 Big picture

- **Question.** How does confirmatory bias - the tendency to interpret ambiguous or conflicting information as supporting one's current hypothesis - distort belief formation?
- **Setting.** The paper studies a simple two-state learning problem. There are two mutually exclusive hypotheses, A and B . A decision maker receives a sequence of informative signals, but sometimes misreads signals that contradict the hypothesis he currently favors.
- **Core behavioral friction.** The agent is otherwise Bayesian, but he is unaware that he sometimes misperceives disconfirming signals. Thus, the belief-updating rule is Bayesian conditional on perceived evidence, while the perception of evidence is biased.
- **Main result 1: overconfidence.** If the agent believes one hypothesis is more likely, a Bayesian observer who understands the bias would generally put less weight on that same hypothesis. The agent's posterior is too extreme.
- **Main result 2: wrongness.** With sufficiently severe confirmatory bias, the agent can believe that one hypothesis is more likely even though an outside Bayesian observer would think the opposite hypothesis is more likely.
- **Main result 3: bad asymptotic learning.** More data do not always fix the problem. If the bias is strong enough, the agent may converge with near certainty to the wrong hypothesis even after receiving infinitely many signals.
- **Economic application.** A principal who knows that an agent is confirmatory-biased may optimally mute incentives to collect information, because more information can make the agent more confidently wrong and induce excessive risky action.
- **Takeaway.** First impressions matter because early beliefs change the way later evidence is perceived. Learning can reinforce false beliefs rather than eliminate them.

2 Incremental contribution of the paper

- **From informal psychology to formal economics.** The paper converts a large psychological literature on confirmatory bias into a tractable economic model of belief formation.
- **A minimal deviation from Bayesian rationality.** The agent is fully Bayesian after perceiving a signal. The only non-Bayesian element is the biased interpretation of signals that conflict with current beliefs.
- **Overconfidence as an endogenous consequence.** The model shows formally why confirmatory bias generates overconfidence: strong beliefs are likely to have been built partly from misread disconfirming evidence.
- **Wrong beliefs can be more likely than correct beliefs.** The paper goes beyond the usual claim that people are too confident. It shows that a biased agent's favored hypothesis can be less likely than the alternative.
- **Learning can fail in the limit.** A fully Bayesian agent eventually learns the true state from infinitely many independent signals. This paper shows that confirmatory bias can break this convergence result.

- **A bridge to incentive design.** The principal-agent illustration shows that biased belief formation changes optimal incentives, even when there is no standard moral-hazard cost of information acquisition.
- **Useful benchmark for later behavioral economics.** The model provides a clean way to ask how belief distortions affect contracts, delegation, persuasion, stereotypes, and investment decisions.

3 Evidence, primitives, and measurement

3.1 Psychological evidence motivating the model

- The paper reviews evidence that people interpret later information through the lens of earlier beliefs.
- In experiments on capital punishment, subjects who initially favored or opposed capital punishment interpreted the same evidence in ways that moved them farther apart.
- In experiments on nuclear technology, pro- and anti-nuclear subjects reacted differently to the same information about nuclear incidents.
- In the Darley and Gross experiment, subjects saw the same child answer the same academic questions, but prior information about socioeconomic background led them to interpret the performance differently.
- The paper distinguishes confirmatory bias from merely ignoring later evidence. The stronger claim is that people may read ambiguous or conflicting evidence as positive evidence for their current view.

Interpretation: The paper does not estimate the bias from data. Instead, it uses psychological findings as a motivation for a deliberately simple theoretical model.

3.2 States, signals, and priors

- There are two states, $x \in \{A, B\}$.
- The prior is symmetric:

$$\Pr(x = A) = \Pr(x = B) = \frac{1}{2}.$$

- In each period $t = 1, 2, 3, \dots$, the agent receives a signal $s_t \in \{a, b\}$.
- Signals are independent and identically distributed conditional on the state:

$$\Pr(s_t = a \mid A) = \Pr(s_t = b \mid B) = \theta, \quad \theta \in (0.5, 1).$$

- Higher θ means signals are more informative.

Interpretation: This is the canonical Bayesian learning environment. Without bias, repeated signals would eventually reveal the true state.

3.3 Perceived signals and confirmatory bias

- The agent does not necessarily perceive s_t correctly. Let $\sigma_t \in \{\alpha, \beta\}$ denote the perceived signal, where α is interpreted as support for A and β as support for B .
- If the signal confirms the agent’s current favored hypothesis, it is perceived correctly.
- If the signal contradicts the agent’s current favored hypothesis, the agent misreads it as confirming evidence with probability $q > 0$.
- q is the key bias parameter. When $q = 0$, the agent is a standard Bayesian. When $q = 1$, the first signal effectively determines all later beliefs because every contradictory signal is reinterpreted as confirmatory.

Interpretation: The paper models confirmatory bias as a perception error, not as a preference for believing one hypothesis.

3.4 Perceived signal probabilities

Suppose the agent currently favors one hypothesis. If the favored hypothesis is actually false, the probability that the agent nevertheless perceives a confirming signal is

$$\theta^* = (1 - \theta) + q\theta. \quad (1)$$

If the favored hypothesis is actually true, the probability that the agent perceives a confirming signal is

$$\theta^{**} = \theta + q(1 - \theta). \quad (2)$$

- θ^* is especially important for wrong learning. If $\theta^* > 1/2$, then once the agent favors the wrong hypothesis, perceived evidence tends to reinforce that wrong hypothesis.
- $\theta^{**} > \theta$ captures the fact that even when the agent is right, confirmatory bias makes supportive evidence appear too frequent.

3.5 Agent posterior and outside observer posterior

If the agent has perceived n_α signals supporting A and n_β supporting B , his likelihood ratio is

$$\Lambda(n_\alpha, n_\beta) = \frac{\Pr(x = A \mid n_\alpha, n_\beta)}{\Pr(x = B \mid n_\alpha, n_\beta)} = \left(\frac{\theta}{1 - \theta} \right)^{n_\alpha - n_\beta}. \quad (3)$$

The authors compare this to an outside Bayesian observer’s likelihood ratio,

$$\Lambda^*(n_\alpha, n_\beta), \quad (4)$$

where the observer knows that the agent suffers from confirmatory bias and observes the number of perceived signals, but not the order in which they arrived.

Interpretation: The gap between Λ and Λ^* is the key diagnostic object. If $\Lambda > \Lambda^*$ when the agent favors A , the agent is overconfident in A .

4 Models and theoretical specifications

4.1 Benchmark Bayesian learning without bias

If $q = 0$, perceived signals are true signals. The agent's posterior follows ordinary Bayes' rule, and with infinitely many independent informative signals the posterior converges to the true state.

Interpretation: The paper's results are not driven by weak information. The environment contains enough information for Bayesian learning to succeed.

4.2 Biased Bayesian learning with confirmatory perception

With $q > 0$, the agent still applies Bayes' rule to perceived evidence. However, the perceived evidence process is endogenous to the current belief. The current favored hypothesis changes the probability distribution of future perceived signals.

The law of perceived signals is therefore history dependent:

- when the agent favors A , signals against A can be turned into perceived signals for A ;
- when the agent favors B , signals against B can be turned into perceived signals for B ;
- when the agent is exactly indifferent, no current hypothesis is favored and signals are perceived correctly.

Interpretation: The model captures a feedback loop: beliefs shape evidence, and evidence then strengthens beliefs.

4.3 Overconfidence result

Proposition 1 states that if the agent perceives more evidence for A than for B , and has received at least one signal, then

$$\Lambda^*(n_\alpha, n_\beta) < \Lambda(n_\alpha, n_\beta). \quad (5)$$

- The agent is overconfident in the hypothesis he currently favors.
- The outside observer discounts the agent's belief because some perceived confirming signals may actually have been disconfirming signals.
- The result holds even though the agent may occasionally have been biased against the hypothesis he now favors when he previously favored the other hypothesis.

Interpretation: The dominant force is that, conditional on currently favoring A , the agent probably spent more time in the past favoring A than favoring B .

4.4 Wrongness result

The paper then shows that overconfidence can become wrongness. With sufficiently severe bias, an outside Bayesian observer may believe the opposite hypothesis is more likely.

Proposition 2 considers the case in which q and θ are both close to 1. If the agent favors A and has perceived one or fewer signals favoring B , then A is probably correct. But if the agent favors

A and has perceived two or more signals favoring B , then the outside observer may conclude that B is more likely.

Interpretation: When the bias is severe, perceived confirming signals after the first one are not very informative because they may be artifacts of bias. Perceived disconfirming signals are more informative because they survived the bias.

4.5 Average correctness

Proposition 3 says that if the observer only knows that the agent favors A , without knowing how strong the agent's belief is, then the observer also favors A :

$$\Lambda^*(n) > 1. \quad (6)$$

- Wrongness is possible for particular belief histories.
- On average, however, the agent's favored hypothesis is still more likely to be true.
- The model therefore distinguishes being too confident from being systematically wrong in the direction of one's best guess.

4.6 Infinite-signal learning

A standard Bayesian eventually learns the truth. The confirmatory agent may not. Let P_W denote the probability that the agent, starting from a neutral prior, eventually believes the wrong hypothesis with near certainty.

Proposition 4 shows that

$$P_W > 0 \quad \text{if} \quad q > 1 - \frac{1}{2\theta}, \quad (7)$$

and

$$P_W = 0 \quad \text{if} \quad q \leq 1 - \frac{1}{2\theta}. \quad (8)$$

Interpretation: If bias is strong enough, then once the agent starts favoring the wrong hypothesis, perceived evidence has positive drift in the wrong direction.

4.7 Incorrect priors

Proposition 5 studies the case in which the agent starts with an incorrect prior belief. For any initial incorrect belief $\mu > 1/2$, the agent can almost surely maintain the wrong belief if either:

- the confirmatory bias q is sufficiently high; or
- true signals are sufficiently weak, with θ close to $1/2$.

Interpretation: The weaker the objective feedback from the environment, the more powerful even a modest confirmatory bias becomes.

4.8 Principal-agent illustration

The final formal section embeds the biased learner in a simple agency problem.

- The principal has wealth $W = 1$.
- The agent can allocate wealth between a safe asset I_C with return 1 and two risky assets I_A and I_B .
- If the state is A , asset I_A pays R and I_B pays 0; if the state is B , asset I_B pays R and I_A pays 0.
- The agent can gather information and then choose the allocation.
- If the agent becomes overconfident from biased learning, he may put all wealth into the wrong risky asset.

Proposition 6 states that if signals are weak or confirmatory bias is severe, the principal may optimally not induce the agent to become informed. Proposition 7 adds that greater risk aversion makes the principal more likely to avoid such incentives.

Interpretation: More information is not always valuable when the agent processes information in a biased way. Incentives to search may generate confident but distorted decisions.

5 Identification and empirical design

5.1 What the model isolates

- The paper isolates one behavioral departure from Bayesian rationality: confirmatory misperception of disconfirming evidence.
- Other biases are deliberately excluded so that the consequences of this one bias can be clearly characterized.
- The model is theoretical rather than empirical. It does not estimate q or θ from data.

Interpretation: The contribution is a mechanism and set of comparative statics, not a causal empirical estimate.

5.2 Key assumptions

- **Two states.** The world is reduced to two exhaustive hypotheses.
- **Symmetric prior.** The baseline prior assigns equal probability to A and B .
- **Independent signals.** Signals are i.i.d. conditional on the true state.
- **Fixed signal precision.** Signal informativeness θ is constant over time.
- **Constant bias.** The probability q of misreading conflicting evidence does not depend on how strong the current belief is.
- **Unawareness.** The agent does not know he is biased, so he does not correct for misperception.

5.3 Why the assumptions are useful

- They make it possible to separate perception bias from Bayesian inference.
- They allow clean comparisons between the agent’s posterior and the posterior of an unbiased Bayesian observer.
- They highlight that even a simple, symmetric, informative environment can generate persistent false certainty.

5.4 Limitations of the model

- The model does not allow the strength of confirmatory bias to rise with belief strength, though that might be psychologically plausible.
- The model assumes the agent misreads contradictory evidence as confirming evidence, whereas in some contexts people may merely ignore contradictory evidence.
- The observer is assumed to know the bias process and the number of perceived signals, but not the order of signals.
- The model has no strategic persuasion, social interaction, or endogenous information source.

Bottom line: The assumptions are intentionally stark. Their purpose is to show that confirmatory bias alone can produce overconfidence, wrongness, and failure of asymptotic learning.

6 Tables and figures

6.1 Table I: Probability of wrongness, $n = 50$, $\eta = 1$

Read:

- Table I reports the probability that the agent is “wrong” after 50 signals under one threshold of wrongness.
- Rows vary the bias parameter q ; columns vary signal precision θ .
- Wrongness is not monotone in q : as q rises, perceived signals become more confirmatory, but mixed-signal histories also become less likely.
- For example, when $\theta = 0.6$ and $q = 0.5$, the table reports a 27 percent probability of wrongness.

Economic message: The table shows that wrongness is not just a logical possibility. Under moderate signal precision and moderate to high bias, the probability can be economically large.

6.2 Table II: Probability of wrongness, $n = 50$, $\eta = 1/2$

Read:

- Table II uses a stricter threshold for wrongness than Table I.
- The probabilities are generally smaller than in Table I, but wrongness remains common for some parameter values.

- With $\theta = 0.6$ and $q = 0.5$, the probability is 19 percent under this more demanding threshold.
- The table reinforces that mistaken beliefs can be meaningfully strong, not just barely on the wrong side of 50 percent.

Economic message: Confirmatory bias can make agents confidently wrong, not merely weakly mistaken.

TABLE I PROBABILITY OF "WRONGNESS," $n = 50, \eta = 1$				
q	θ			
	0.6	0.7	0.7	0.9
.1	—	—	—	—
.2	12	2	0	—
.3	21	9	1	0
.4	29	15	5	1
.5	27	18	10	3
.6	33	22	12	5
.7	27	21	15	7
.8	33	24	15	8
.9	21	17	12	9

(a) Table I: wrongness after 50 signals, threshold $\eta = 1$

TABLE II PROBABILITY OF "WRONGNESS," $n = 50, \eta = 1/2$				
q	θ			
	0.6	0.7	0.7	0.9
.1	—	—	—	—
.2	—	—	—	—
.3	10	5	1	—
.4	15	13	5	1
.5	19	18	9	3
.6	16	18	12	5
.7	18	21	13	7
.8	11	16	15	7
.9	4	8	12	6

(b) Table II: wrongness after 50 signals, threshold $\eta = 1/2$

6.3 Table III: Probability of wrongness, $n = 50, \eta = 1/9$

Read:

- Table III imposes an even stronger wrongness threshold.
- The agent must be wrong by a larger margin for the event to count.
- Probabilities fall relative to Tables I and II, but are still positive in many cases.
- The strongest wrongness probabilities appear when signals are not too precise and bias is substantial.

Economic message: Severe mistakes are less common than moderate mistakes, but the model predicts that they can still occur with nontrivial probability.

6.4 Table IV: Probability of wrongness, $n = 7, \eta = 1$

Read:

- Table IV repeats the wrongness calculation with only 7 signals.
- Wrongness can arise even with a small sample of information.
- With fewer signals, probabilities are sensitive to discrete signal histories.
- The table links the numerical examples in the text to finite-sample belief mistakes.

Economic message: First impressions matter early. A small number of signals can put the agent on a path where later evidence is processed through a distorted lens.

TABLE III
PROBABILITY OF "WRONGNESS," $n = 50$, $\eta = 1/9$

q	θ			
	0.6	0.7	0.7	0.9
.1	—	—	—	—
.2	—	—	—	—
.3	—	—	—	—
.4	—	5	—	—
.5	3	12	7	1
.6	3	14	11	4
.7	2	11	12	6
.8	1	6	12	7
.9	0	4	7	6

(a) Table III: wrongness after 50 signals, threshold $\eta = 1/9$

TABLE IV
PROBABILITY OF "WRONGNESS," $n = 7$, $\eta = 1$

q	θ			
	0.6	0.7	0.7	0.9
.1	—	—	—	—
.2	—	—	—	—
.3	—	—	—	—
.4	—	—	—	—
.5	—	—	—	2
.6	—	5	3	1
.7	3	3	2	5
.8	10	8	6	3
.9	3	2	2	1

(b) Table IV: wrongness after 7 signals, threshold $\eta = 1$

6.5 Table V: Probability of believing in the wrong hypothesis after infinitely many signals

Read:

- Table V reports P_W , the probability that the agent converges to the wrong hypothesis after infinitely many signals.
- When q is too low relative to θ , the probability is zero: the agent eventually corrects mistakes.
- Once $q > 1 - 1/(2\theta)$, P_W becomes positive.
- For $q = 0.5$ and $\theta = 0.75$, the paper reports $P_W = 7/52$, about 13 percent.
- For weaker signals, such as $\theta = 0.6$, the probability can be much larger.

Economic message: More data do not guarantee truth. If confirmatory bias is strong enough, infinite evidence can produce false certainty.

6.6 Table VI: Probability of maintaining an incorrect prior belief after infinitely many signals

Read:

- Table VI starts the agent with an incorrect prior belief and asks whether the wrong belief survives forever.
- The table varies q , θ , and the strength of the incorrect prior μ .
- A stronger initial wrong belief is represented by a larger $D(\mu)$, the number of net contrary perceived signals needed to return to indifference.
- Even modest bias can make incorrect priors highly persistent when signals are weak.
- For example, when $\theta = 0.551$, $\mu = 0.6$, and $q = 0.333$, the table reports a 79 percent probability of maintaining the wrong belief.

Economic message: Initial errors are especially dangerous when feedback is noisy. The model explains why early stereotypes, first impressions, or initial theories can become self-reinforcing.

TABLE V
PROBABILITY OF BELIEVING IN WRONG HYPOTHESIS AFTER OBSERVING AN INFINITE
NUMBER OF SIGNALS

q	$\theta = .6$	$\theta = .667$	$\theta = .75$	$\theta = .9$
.25	18	—	—	—
.333	26	12	—	—
.5	34	24	13	2
.75	38	31	22	8

(a) Table V: convergence to the wrong hypothesis

TABLE VI
PROBABILITY OF MAINTAINING AN INCORRECT PRIOR BELIEF AFTER RECEIVING AN
INFINITE NUMBER OF SIGNALS

q	$\theta = .6, \mu = .6, D(\mu) = 1$	$\theta = .551, \mu = .6, D(\mu) = 2$	$\theta = .75, \mu = .75, D(\mu) = 1$	$\theta = .634, \mu = .75, D(\mu) = 2$
.25	32	67	—	24
.33	51	79	—	56
.5	72	92	48	85
.75	89	99	82	98

(b) Table VI: persistence of an incorrect prior

7 Short summary

- The paper builds a formal model of confirmatory bias.
- The agent chooses between two hypotheses and receives informative signals.
- The agent sometimes misreads signals that contradict his current favored hypothesis.
- The agent updates rationally from perceived signals but does not account for misperception.
- The main comparison is between the agent’s posterior and the posterior of an unbiased Bayesian observer who understands the bias.
- Confirmatory bias generally produces overconfidence in the favored hypothesis.
- If the bias is severe, the agent’s favored hypothesis can be less likely than the alternative.
- Infinite information does not always solve the problem; the agent may converge to a false hypothesis.
- Incorrect priors are especially persistent when signals are weak or the bias is strong.
- In a principal-agent setting, the principal may optimally avoid inducing a confirmatory agent to gather more information.

8 Conclusion

8.1 Core conclusion

Confirmatory bias can transform ordinary Bayesian learning into a process that amplifies first impressions. Because later evidence is interpreted through current beliefs, perceived evidence becomes endogenous to the belief history. This makes posterior beliefs too extreme and can make false beliefs self-reinforcing.

8.2 Economic intuition

A strong belief is not only a signal that the agent has seen much supportive evidence. It is also a warning sign that the agent may have spent many periods biased toward interpreting evidence as supportive. An outside observer should therefore discount the strength of the agent’s conviction.

8.3 Incremental contribution revisited

The paper’s main contribution is to show that a single, clearly modeled psychological bias can overturn several standard implications of Bayesian learning. It generates overconfidence, wrongness, and failure of asymptotic learning in an otherwise simple information environment.

8.4 Implications for economics

The mechanism has implications for financial decisions, managerial delegation, expert advice, stereotypes, political beliefs, scientific disagreement, and organizational learning. In each case, incentives to acquire more information may not be enough if the decision maker processes information in a biased way.

8.5 Limitations

The model is deliberately stylized. It has two states, fixed signal quality, a constant bias parameter, no strategic communication, and no empirical estimation of the bias. These simplifications make the results transparent, but applications require care.

8.6 Takeaway

First impressions matter because they shape the interpretation of future evidence. In environments where evidence is ambiguous or weak, confirmatory bias can make people increasingly confident in views that an unbiased observer would reject.